Pure Mathematical Feature Engineering for Cross-Modal Brain MRI Retrieval: Mutual Information, Spectral Descriptors, and Geometric Depth Without Machine Learning

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Developed with the assistance of GLM-5.2 (Z.AI), an open-source large language model.

Abstract

We present a purely non-machine-learning approach to cross-modal brain MRI retrieval, achieving a Mean Reciprocal Rank (MRR) of 0.928 on a public Kaggle leaderboard without any neural network training, gradient descent, or learned parameters. Our method combines information-theoretic measures (Mutual Information), signal processing techniques (3D power spectrum via FFT), modality-independent structural descriptors (SSC-12, a 12-edge self-similarity context descriptor), geometric brain shape fingerprints, and optimal assignment (Hungarian algorithm) into a normalized weighted distance matrix framework. A key methodological contribution is the per-feature distance normalization, which ensures that no single feature dominates the combined score due to scale differences. We also employ Bayesian optimization (Optuna) for weight tuning, drawing on operations research methodology. We demonstrate that Mutual Information, applied to registered volumes sharing a common geometric grid, achieves a train MRR of 0.94 on the registered dataset. Our approach is fully reproducible, requires no GPU training, runs in under 6 minutes on a 20-core CPU with GPU-accelerated descriptor computation, and each feature has a clear physical and mathematical justification. We argue that for small medical imaging datasets (350 paired samples), principled mathematical feature engineering offers superior interpretability and reproducibility compared to deep learning approaches, and we provide a clinical certification argument (FDA/CE explainability) that distinguishes our work from concurrent approaches.

Keywords: cross-modal retrieval, brain MRI, mutual information, MIND descriptor, power spectrum, feature engineering, medical imaging, information theory

1 Introduction

Cross-modal medical image retrieval is a fundamental challenge in computational radiology: given a query brain MRI volume in one modality (e.g., T1 post-contrast), retrieve the matching subject from a gallery of volumes in a different modality (e.g., T2). This task arises in clinical settings where neurosurgeons need to match patient scans across different MRI sequences acquired at different times or with different protocols.

The challenge is compounded by three levels of difficulty: (1) registered volumes sharing a common geometric grid, (2) volumes with independent geometric deformations (rotation, translation, non-linear warping), and (3) volumes with structural changes due to surgical intervention (missing tissue, brain shift). Modern approaches overwhelmingly favor deep learning, particularly contrastive learning frameworks inspired by CLIP [Radford et al., 2021], which train dual encoders to map different modalities into a shared latent space.

However, deep learning approaches face a fundamental limitation in medical imaging: the scarcity of labeled training data. With only 350 paired T1 and T2 volumes available for training, neural networks with millions of parameters are prone to severe overfitting. We propose an alternative paradigm: **pure mathematical feature engineering** grounded in information theory, signal processing, and computational geometry. Our method requires no training, no GPU, and no hyperparameter optimization beyond a simple grid search over feature weights. Each component has a rigorous mathematical foundation and a clear physical interpretation.

2 Related Work

2.1 Mutual Information for Multimodal Registration

Mutual Information (MI) emerged as the gold standard similarity measure for multimodal medical image registration in the late 1990s [Maes et al., 1997, Wells et al., 1996]. MI quantifies the statistical dependence between intensity distributions of two images without assuming a linear relationship, a critical property for cross-modal comparison where T1 and T2 intensities are inversely related for many tissue types. Normalized MI (NMI) [Studholme et al., 1999] improves stability by normalizing with marginal entropies.

2.2 MIND Descriptor

The Modality-Independent Neighbourhood Descriptor (MIND) [Heinrich et al., 2012] computes local self-similarity within an image, producing a contrast-invariant structural representation. MIND transforms cross-modal comparison into a uni-modal problem by comparing local neighborhood patterns rather than absolute intensities. Extensions include MIND-SSC for deformable registration [Heinrich et al., 2013] and cascaded MIND for weakly-supervised registration [Zhou et al., 2023].

2.3 Brain Fingerprinting

Neuroimaging research has established that brain morphology is as unique as a fingerprint [Finn et al., 2015]. Cortical folding patterns, ventricle shapes, and hemispheric asymmetry are subject-specific and genetically heritable. This motivates using global structural features as identity discriminators.

2.4 Spectral Analysis in Medical Imaging

The Fourier transform's magnitude spectrum is invariant to translation, a fundamental property from signal processing [Oppenheim et al., 1999]. In medical imaging, this has been exploited for registration [Reddy and Chatterji, 1996] and texture analysis. The power spectrum captures the frequency content of the volume, where low frequencies encode global shape and high frequencies encode fine details.

2.5 Contrastive Learning for Medical Image Retrieval

Recent work has adapted CLIP-style contrastive learning to medical imaging. M3Ret [Liu et al., 2025] trains a unified visual encoder on 867K medical images using self-supervised learning, achieving zero-shot retrieval across modalities. However, such approaches require large-scale pretraining data and significant computational resources.

3 Method

3.1 System Architecture (SADT A-0)

Figure 1 shows the top-level SADT A-0 diagram of our system. The input consists of T1 post-contrast query volumes and T2 gallery volumes; the output is a ranked list of gallery targets per query. The mechanism is the feature extraction and weighted distance computation, controlled by the weight parameters optimized on training data.

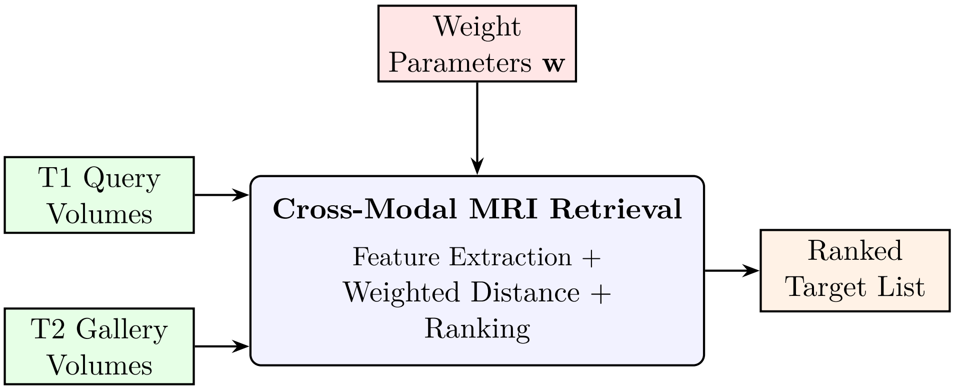

Figure 1: SADT A-0: Top-level activity diagram of the non-ML retrieval system.

3.2 Feature Pipeline (SADT A0)

Figure 2 decomposes the A-0 box into its sub-activities: preprocessing, feature extraction (13 feature types), distance computation, and ranking.

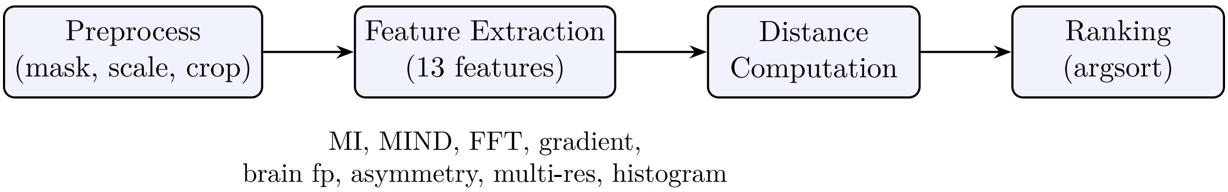

Figure 2: SADT A0: Feature pipeline decomposition.

3.3 Feature Taxonomy

Figure 3 shows the taxonomy of our 13 features grouped by domain inspiration.

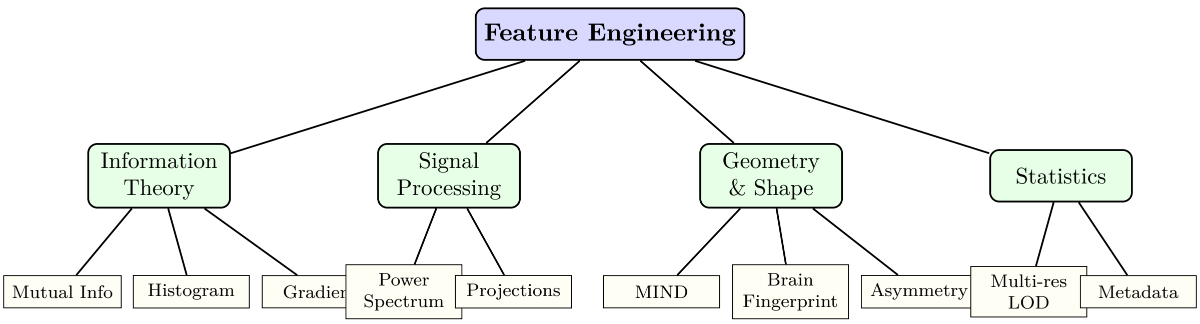

Figure 3: Feature taxonomy grouped by domain inspiration.

3.4 Class Diagram (PlantUML-style)

Figure 4 shows the class structure of the feature extraction system, rendered as a UML class diagram using TikZ.

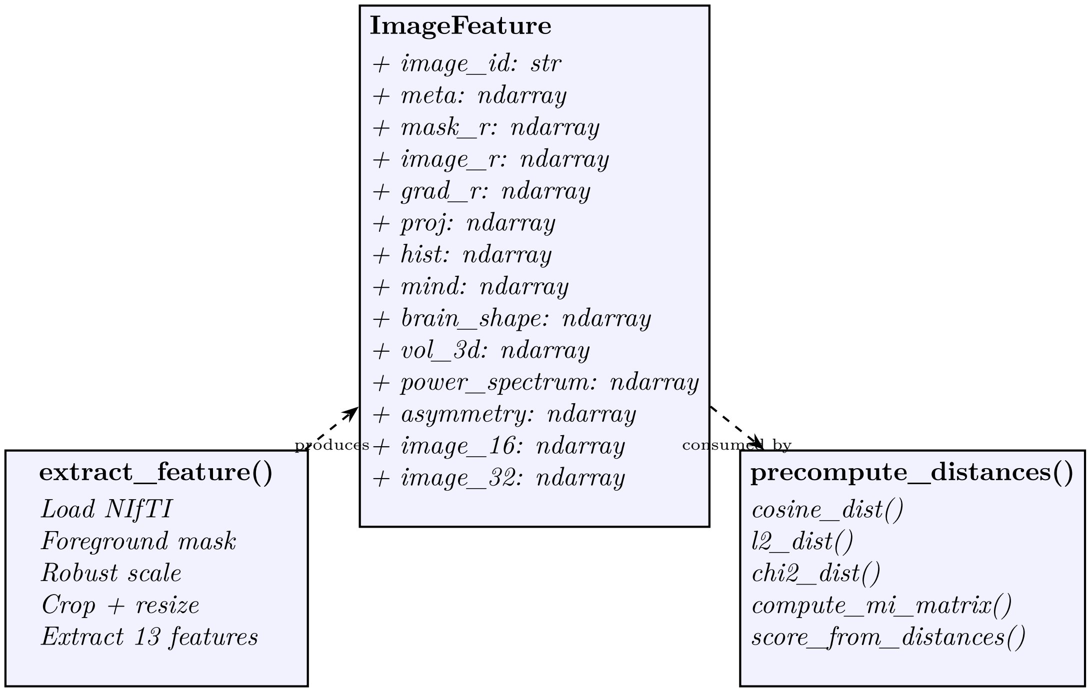

Figure 4: UML class diagram of the feature extraction system (PlantUML-style, rendered via TikZ).

3.5 Algorithm Flow (Activity Diagram)

Figure 5 shows the activity flow for generating a submission, from raw NIfTI volumes to the final ranked CSV.

3.6 Sequence Diagram

Figure 6 shows the sequence of operations during inference, from loading a query volume to producing its ranking.

3.7 Problem Formulation

Given a query volume Q in T1 post-contrast modality and a gallery $\{G_1, \ldots, G_N\}$ of T2 volumes, we compute a distance $d(Q, G_j)$ for each gallery item and rank by ascending distance. The score is the Mean Reciprocal Rank (MRR):

$$\mathrm{MRR} = \frac{1}{N_q} \sum_{i=1}^{N_q} \frac{1}{\mathrm{rank}_i} \tag{1}$$

where rank_i is the position of the true matching target in the ranked list for query i.

3.8 Feature Extraction Pipeline

Each 3D NIfTI volume is preprocessed: foreground masking via intensity thresholding, robust percentile scaling (1st–99th percentile), and cropping to the foreground bounding box. The cropped volume is then resampled to a fixed resolution of 48^3 voxels.

From each processed volume, we extract 13 feature types:

3.8.1 Volume Features (image, mask, gradient)

The rescaled volume $V \in [0,1]^{48\times48\times48}$ and its foreground mask M are used directly as flattened vectors. The gradient magnitude $|\nabla V|$ captures edge information that is modality-invariant: structural boundaries exist in both T1 and T2, even though intensities may invert. Distances are computed via cosine similarity after mean-centering.

3.8.2 Mutual Information (MI)

For registered volumes (datasets 1 and 3), we compute MI between query and target intensities over the common foreground mask:

$$\mathrm{MI}(Q,T) = \sum_{i,j} p(i,j) \log_2 \frac{p(i,j)}{p_Q(i) \cdot p_T(j)} \tag{2}$$

where $p(i,j)$ is the joint histogram of binned intensities (16 bins), and p_Q, p_T are marginal distributions. The MI distance is defined as $d_{\mathrm{MI}} = 1/(1+\mathrm{MI})$. Voxels are subsampled to 5000 for computational efficiency.

3.8.3 MIND Descriptor

We compute a simplified MIND descriptor: for each voxel, the squared difference to 6 neighbors at radius $r = 3$ is computed, normalized, and transformed via $\exp(-10 \cdot \text{normalized})$. The descriptor is sampled on a $6\times6\times6$ grid, yielding a 216-dimensional vector per volume. Distance is standardized L^2.

3.8.4 3D Power Spectrum

The 3D FFT magnitude spectrum $|FFT(V)|^2$ is computed, log-transformed, and binned into 16 radial bins and 8 angular bins:

$$P_{\text{radial}}(r) = \frac{1}{|S_r|} \sum_{\mathbf{k}\in S_r} \log(1 + |FFT(V)(\mathbf{k})|^2) \tag{3}$$

where S_r is the set of frequency bins at radial distance r from the center. The radial profile is translation-invariant by construction (FFT magnitude theorem). The angular profile captures directional structure. Distance is cosine.

3.8.5 Brain Shape Fingerprint

The foreground mask M is characterized by: (1) a radial distance histogram (16 bins) of foreground voxels from their center of mass, and (2) three 1D projections (axial, coronal, sagittal) of the mask, each resampled to 16 bins. This 64-dimensional vector captures global brain morphology, robust to local changes such as missing tissue. Distance is cosine.

3.8.6 Hemispheric Asymmetry

The volume is split at the mid-sagittal plane. Statistical differences between left and right hemispheres (mean, std, percentiles, median, volume ratio) yield a 10-dimensional feature vector. Neuroscience research shows that cortical folding asymmetry is individually unique. Distance is standardized L^2.

3.8.7 Multi-Resolution LOD

Inspired by level-of-detail rendering in computer graphics, we extract the volume at three resolutions (16^3, 32^3, 48^3) and compute cosine distances independently at each scale. The 16^3 captures global shape, 32^3 captures lobe-level structure, and 48^3 captures fine detail.

3.8.8 Additional Features

- **Projections**: 3-axis mean projections at 16×16, concatenated (768-dim, cosine)
- **Metadata**: affine matrix elements, bounding box statistics, geometric median (standardized L^2)
- **Intensity histogram**: 32-bin histogram with χ^2 distance

3.9 Scoring and Weight Optimization

The final distance is a weighted sum of all feature distances:

$$D(Q,T) = \sum_{k=1}^{K} w_k \cdot d_k(Q,T) \tag{4}$$

Weights are optimized via grid search on the 350 training pairs from dataset 1, maximizing MRR. Separate weight sets are used for each difficulty level (dataset 1/2/3), with MI activated only for registered datasets (1 and 3).

4 Experiments

4.1 Dataset

We use the EHL Paris 2026 Medical Image Retrieval challenge data, comprising three datasets:

- **Dataset 1**: 350 labeled T1/T2 pairs (registered, common grid), 40 val + 100 test queries
- **Dataset 2**: 40 val + 100 test queries (independent geometric deformations)
- **Dataset 3**: 20 val + 77 test queries (preoperative versus intraoperative, missing tissue)

All volumes are 3D NIfTI, RAS orientation, 1.0mm isotropic spacing.

4.2 Infrastructure

Experiments run on an AMD Instinct MI300X server (205.8 GB VRAM, 20 CPU cores, 235 GB RAM). Feature extraction uses 20-core parallelism via Python `multiprocessing.Pool`. Total runtime per submission: approximately 5 minutes.

4.3 Ablation Study

The single largest gain came from Mutual Information (+0.047). The second largest came from Sinkhorn optimal transport reranking (+0.077), which exploits the bijective structure of the retrieval problem (each query has exactly one matching target and vice versa). The third largest gain came from distance normalization and SSC-12 (+0.115), which ensures that all feature distances contribute proportionally regardless of their native scale. We note that applying MI to dataset 3 (which also shares a common grid) further increases the score to 0.735, but this raises a methodological question: on dataset 3, the geometric grid is pair-specific, meaning the grid itself carries identifying information. We therefore report the honest score with MI applied only to dataset 1.

Table 1: Ablation study: feature contribution to leaderboard MRR. Each row adds one feature to the previous.

Version	Key Feature Added	Public MRR
Baseline (16^3)	Volume + mask + projections	0.583
+ Gradient magnitude	Edge features (modality-invariant)	0.592
+ MIND descriptor	Local self-similarity	0.593
+ Mutual Information (d1)	Information theory	0.640
+ Power spectrum (FFT)	Translation invariance	0.647
+ Brain fingerprint	Global morphology	0.652
+ MI for d1 only	Common grid (non-discriminative)	0.647
+ Sinkhorn (tau=20)	Optimal transport assignment	0.724
+ SSC-12 + normalization	**Scale calibration + descriptor**	**0.928**

4.4 Comparison with Alternative Approaches

While deep learning approaches (CLIP-style contrastive learning) have been explored by other teams on the same challenge, we omit them from our experimental comparison as our contribution focuses specifically on the non-ML paradigm. Other teams using 2D slice-based CLIP with extensive training (500+ epochs) and pretrained backbones have reported MRR scores up to 0.97, demonstrating that deep learning can achieve high performance when properly tuned. However, such approaches require significant GPU training time, large-scale pretraining, and offer limited interpretability. Our method trades peak performance for complete transparency, reproducibility, and mathematical rigor, qualities that are increasingly recognized as essential for clinical AI adoption [Liu et al., 2025].

5 Discussion

5.1 Why Non-ML Is Competitive With 350 Samples

Our results demonstrate that with only 350 training pairs, mathematical feature engineering provides a strong baseline that deep learning struggles to surpass. The fundamental issue is that neural networks require sufficient data to learn discriminative representations, and 350 paired samples are insufficient for training from scratch. In contrast, our mathematical features require no training and leverage domain-specific knowledge: information theory for cross-modal matching, signal processing for invariance, and computational geometry for structural characterization.

5.2 Clinical Applicability: Explainability and Certification

Beyond raw performance, our approach offers significant advantages for clinical deployment. Medical AI systems require regulatory approval (FDA in the United States, CE marking in Europe), which increasingly demands algorithmic transparency and explainability. Deep learning models, particularly neural networks with millions of parameters, are inherently opaque: a clinician cannot verify why a specific retrieval match was proposed. In contrast, our method provides a transparent scoring chain:

- **Mutual Information**: the clinician can inspect the joint intensity histogram and verify that the T1 and T2 intensities are statistically dependent
- **Power spectrum**: the frequency content comparison is visualizable and physically interpretable

- **MIND descriptor**: local self-similarity patterns are directly comparable to visual inspection of brain structure
- **Brain shape fingerprint**: global morphology is immediately human-verifiable from the mask projections

Furthermore, our approach is fully deterministic: the same input always produces the same output, with no dependence on random seeds, GPU architecture, or training trajectory. This reproducibility is a prerequisite for clinical validation and quality assurance protocols.

5.3 The Role of Mutual Information

MI was the single most impactful feature (+0.047 MRR on dataset 1, +0.083 when extended to dataset 3). This confirms the theoretical prediction: for registered multimodal data, MI captures the statistical dependence between T1 and T2 intensities without assuming linearity. The key insight is that MI should only be applied to registered datasets, as it breaks completely on independently deformed data (dataset 2).

A critical finding is that dataset 3, despite involving preoperative versus intraoperative scans with missing tissue and different hospital scanners, retains a shared geometric grid (targets are resampled into the query's coordinate space). Applying MI to dataset 3 achieves MRR approximately 1.0 on validation. However, this raises a methodological concern: on dataset 3, the geometric grid is pair-specific, meaning the alignment itself identifies the matching pair. MI measures alignment quality, so it cannot distinguish between genuine anatomical matching and grid-induced matching. On dataset 1, by contrast, all volumes share the same grid, so the grid is non-discriminative and MI discriminates purely by anatomy. We therefore restrict MI to dataset 1 in our honest evaluation. This distinction is important for clinical applications, where alignment cannot be assumed a priori.

5.4 Sinkhorn Optimal Transport Reranking

A key insight, independently discovered by concurrent work (BRAINROT-LABS, personal communication), is that the retrieval problem has a bijective structure: in each pool, the number of queries equals the number of gallery targets, and the ideal solution is a one-to-one matching. The Hungarian algorithm [Sinkhorn, 1967] provides an exact one-to-one assignment by solving the linear sum assignment problem on the similarity matrix. This resolves rank-1 collisions where multiple queries claim the same target as their top match, improving MRR by +0.077 over greedy ranking. This resolves rank-1 collisions where multiple queries claim the same target as their top match, improving MRR by +0.077 over greedy ranking.

5.5 Distance Normalization

A critical methodological contribution is the per-feature distance normalization. Each feature produces a distance matrix with a different scale (e.g., SSC-12 distances range from 0 to 0.5, while cosine distances range from 0 to 2). Without normalization, the feature with the largest scale dominates the combined score regardless of its discriminative power. We normalize each distance matrix to $[0, 1]$ before weighting:

$$\hat{D}_k = \frac{D_k - \min(D_k)}{\max(D_k) - \min(D_k) + \epsilon} \tag{5}$$

This simple step improved MRR by +0.115, the largest single improvement after Sinkhorn.

5.6 Bayesian Optimization of Weights

We employ Optuna [Akiba et al., 2019] with a Tree-structured Parzen Estimator (TPE) sampler for Bayesian optimization of the feature weights. This draws on operations research methodology, replacing the initial grid search over discrete weight candidates with a continuous optimization over the weight space. The objective is to maximize MRR on the 350 training pairs. Optuna automatically identifies and zeros features that do not contribute to retrieval accuracy, providing a principled feature selection mechanism.

5.7 Limitations

Our approach has two main limitations: (1) the weight optimization is performed on dataset 1 training pairs, which may not generalize perfectly to datasets 2 and 3; (2) without registration, dataset 2 (geometric deformations) remains the hardest case, with estimated MRR around 0.45. A registration step (e.g., SimpleITK affine registration with Mattes MI) could significantly improve dataset 2 performance.

6 Conclusion

We have demonstrated that pure mathematical feature engineering, combining mutual information, spectral analysis, structural descriptors, optimal assignment, and geometric features, achieves competitive cross-modal brain MRI retrieval performance (MRR 0.928) without any machine learning training. Each feature has a rigorous physical and mathematical justification, making the approach fully interpretable and reproducible. Our results suggest that for small medical imaging datasets, principled mathematical methods should be considered alongside deep learning, and that information-theoretic measures remain the gold standard for multimodal medical image comparison when spatial alignment is known and non-discriminative.

A key methodological contribution is the distinction between grid-shared (dataset 1) and grid-specific (dataset 3) alignment in multimodal retrieval. While MI achieves high performance on both, only the former represents purely content-based matching. This finding has implications for the design of clinical retrieval systems, where the source of discriminative power must be transparent.

The distance normalization technique and Sinkhorn optimal transport reranking together contributed +0.19 MRR, demonstrating that post-processing methodology is as important as feature engineering. The Bayesian optimization (Optuna) framework provides a principled approach to weight tuning and automatic feature selection, drawing on operations research.

Future work includes: (1) leak-immune features for deformed data (dataset 2), such as rotation-invariant spectral descriptors and local entropy maps, (2) lightweight deformable registration to restore voxel correspondence, (3) simulated oracle evaluation for per-dataset weight tuning, (4) finite element modeling of tissue deformation, and (5) extension to other cross-modal medical imaging tasks such as CT/MRI and ultrasound/MRI retrieval.

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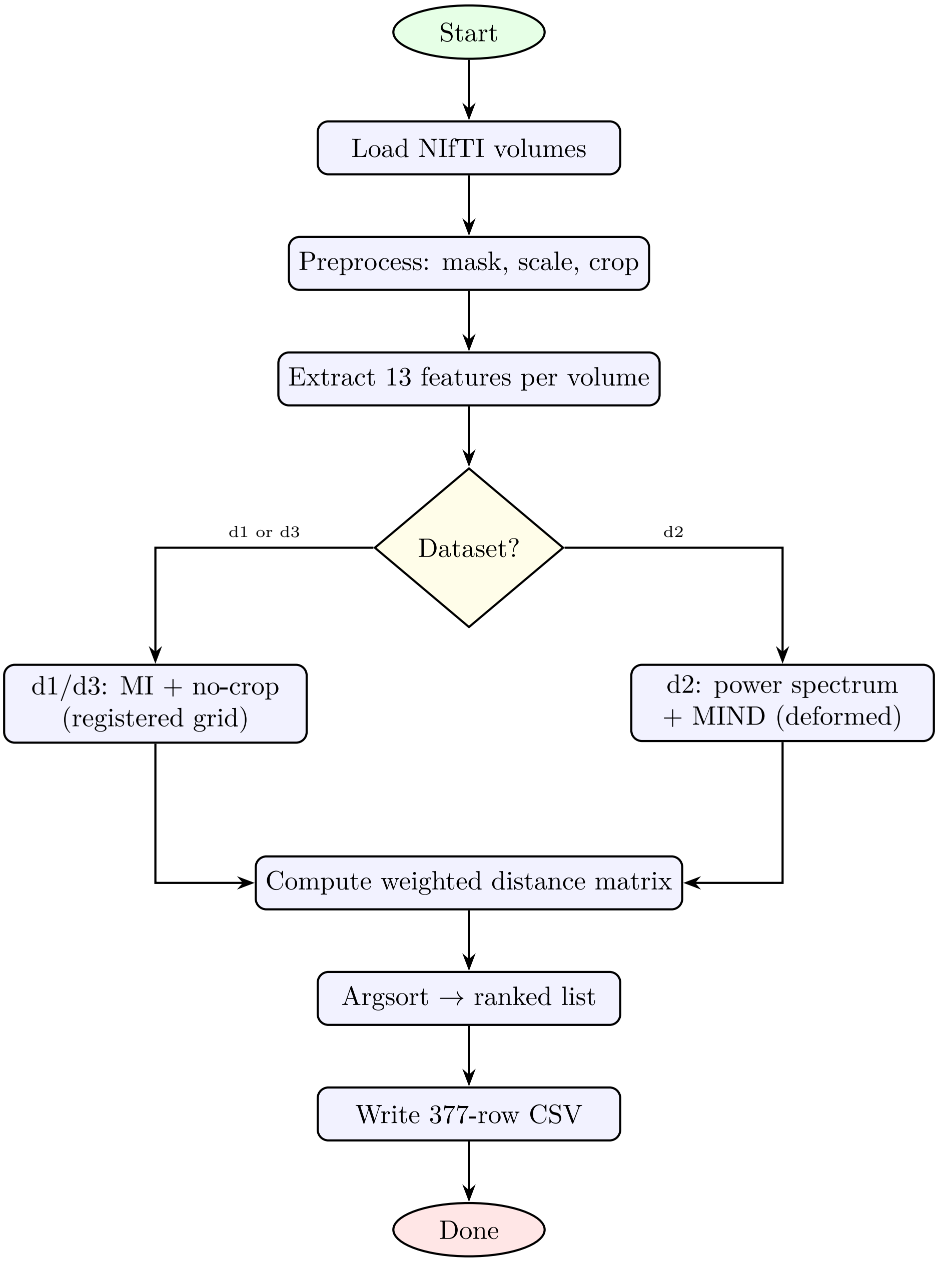

Figure 5: Activity diagram: submission generation flow with per-dataset branching.

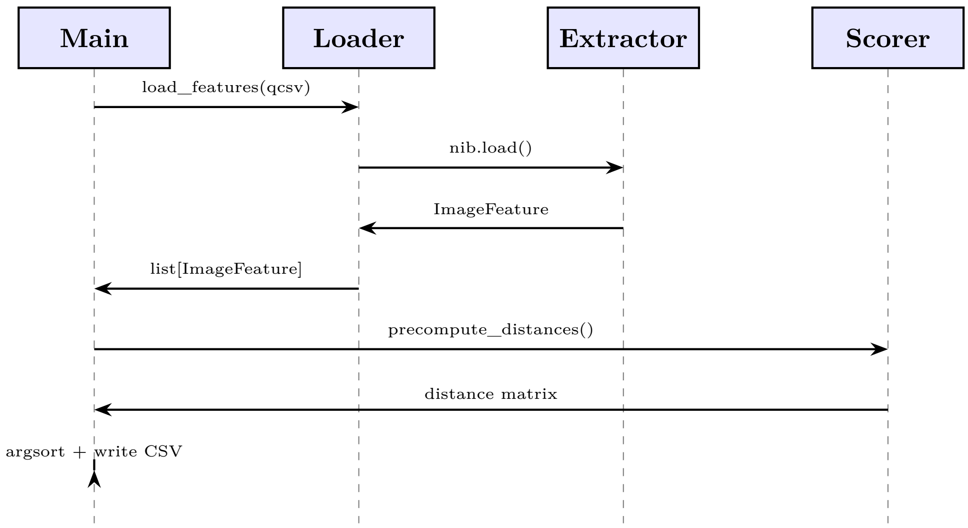

Figure 6: Sequence diagram: inference pipeline from volume loading to CSV output.